

TrustEpisode: Deterministic Episode Formation and Manifest-Scoped Calibration for Partial EDR/NDR Telemetry

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ABSTRACT

Multi-step attacks appear as temporally dispersed endpoint and network observations rather than one decisive alert. Correlation must therefore define which events form a candidate episode, preserve deterministic revisions under late data, and distinguish an unsupported probability from a low probability when telemetry is missing.

We present TRUSTEPISODE, a label-blind EDR/NDR pipeline with explicit ownership boundaries. Evidence Fabric emits canonical events and independent source-health intervals; Episode Synthesis alone owns membership and immutable revisions; a two-feature stateless scorer emits a raw logit; and a manifest-scoped affine calibrator emits a nullable maliciousness probability. A separate deterministic evaluator reports a five-axis evidence-sufficiency profile without changing that probability. An AuditRecord Assembler creates one canonical, append-only record per revision from the independently produced revision, score, support state, probability, sufficiency profile, and evidence references.

We specify a frozen controlled-laboratory protocol for formation, ranking, calibration, partial-telemetry replay, physical collector outage, and deterministic replay. The protocol fixes candidate populations, split assignment, thresholds, group-calibration gates, interventions, matching, metrics, resampling, and empty RT1–RT7 result schemas before locked test access. Empirical result cells remain pending; no synthetic performance values or confidence intervals are reported.

KEYWORDS

attack episode formation, probability calibration, partial telemetry, provenance, EDR, NDR

1 INTRODUCTION

Multi-step attacks rarely produce one conclusive alert. A remote connection, script execution, file transfer, or administrative action can be benign alone yet material when connected by time, entity, session, and process relations. Endpoint detection and response (EDR) and network detection and response (NDR) also fail differently: an event may be absent because no action was observed, because no detector fired, or because a collector or parser was unavailable. Treating all three states as zero evidence silently changes both correlation and probability semantics.

Existing alert correlation and provenance systems reconstruct causal context, but their candidate boundaries, late-data behavior, and calibration contracts are often not explicit enough for byte-identical replay [2, 7–9]. Conversely, an anomaly or severity score is not automatically a probability, and calibration is meaningful only for the candidate population and pipeline version on which it was validated [4, 5]. These gaps are coupled: changing episode membership changes the scored population, while hiding source

failure behind a zero feature can make an unsupported mapping appear confidently benign.

TRUSTEPISODE addresses this problem with a deliberately narrow contract. It transforms only EDR/NDR observations into deterministic EpisodeRevisions, computes a compact raw logit from detector deviation and independent corroboration, emits a calibrated probability only under an exact manifest match, reports evidence sufficiency independently, and assembles one canonical AuditRecord per revision. Ground truth is unavailable to every online producer and is joined only after inference.

Contributions. This paper makes four contributions:

- (1) a single-owner online object model and deterministic episode state machine with fixed guards, total ordering, watermark, and revision policy;
- (2) a compact scorer plus manifest-scoped affine calibration that returns canonical null outside validated support;
- (3) explicit detector-availability and five-axis sufficiency truth tables, together with a machine-readable AuditRecord contract; and
- (4) a preregistered controlled-lab protocol with fixed populations, splits, thresholds, perturbations, matching, statistics, and RT1–RT7 result schemas.

2 RELATED WORK

Alert correlation and provenance. Classical alert-correlation methods connect observations into attack scenarios using temporal, causal, and prerequisite relations [9]. Provenance-based intrusion detection preserves process and information-flow context; HOLMES, NODLINK, and KAIROS show how causal graphs support multi-stage reconstruction and investigation [2, 7, 8]. CAPTAIN further demonstrates that a lightweight, interpretable rule system can adapt deployment-specific parameters [13]. TRUSTEPISODE differs by treating deterministic cross-source candidate construction, revision ownership, and candidate-conditioned probability calibration as one versioned contract rather than presenting a new graph detector.

Calibration and incomplete telemetry. Ranking and calibration answer different questions: a high AUPRC does not imply reliable probabilities [4, 5]. Platt-style affine calibration is compact and ranking-preserving when its slope is positive, but only for a fixed candidate population and support regime [11]. Missing-modality research likewise shows that absent inputs require explicit handling rather than silent zero imputation [14]. TRUSTEPISODE therefore separates observed source masks, authenticated health, detector availability, calibration status, and sufficiency. The raw scorer cannot read health or labels, and unsupported records retain a diagnostic logit but no numeric probability.

Scope. The method does not model operational impact, response priority, autonomous response, CTI enrichment, or language-agent behavior. Its claims are limited to deterministic EDR/NDR episode formation, manifest-scoped maliciousness calibration, explicit evidence sufficiency, and replayable audit records.

3 TRUSTEPISODE METHOD

TRUSTEPISODE converts EDR/NDR telemetry into deterministic candidate revisions and version-scoped evidence records. Evidence Fabric and Health Monitor respectively own events and source health; Episode Synthesis alone owns membership; Feature Extractor emits D, C and detector state; Stateless Scorer emits z ; Scope Resolver emits κ ; Calibrator emits nullable P ; and Sufficiency Evaluator emits structured U . An AuditRecord Assembler, not the store or Episode Synthesis, is the sole producer of one canonical AuditRecord per revision. Ground truth is joined only after inference. The method estimates malicious activity inside a frozen EDR/NDR candidate; it does not estimate enterprise risk, impact, response priority, or unobservable attacks. Figure 1 shows these boundaries.

3.1 Problem Scope, Output Contract, and Ownership

Let C contain every emitted EpisodeRevision and define

$$C_{\text{sup}} = \{\epsilon^{(r)} \in C : \kappa_{\epsilon^{(r)}} \in \{\kappa_G, \kappa_g\}\}, \quad (1)$$

where κ_G, κ_g , and $\kappa_\emptyset$ serialize as `supported_global`, `supported_group`, and `out_of_support`. The offline target $Y_{\epsilon^{(r)}} \in \{0, 1\}$ is revision specific: evidence first accepted in $r+1$ cannot relabel r . Positive and negative revisions form the primary supervised cohort; mixed and ambiguous revisions enter only the registered sensitivity analysis.

When defined, the candidate-conditioned output is

$$P_{\epsilon^{(r)}} = \Pr(Y_{\epsilon^{(r)}} = 1 \mid \epsilon^{(r)}, v_{\text{episode}}, v_{\text{feature}}, v_{\text{model}}, v_{\text{cal}}). \quad (2)$$

It is not the unconditional probability that an organization is under attack. Online schemas are label blind and reject scenario, split, marker, window, and label fields. Collector identity and health messages are authenticated; activity outside observable EDR/NDR traces is outside the claim.

3.2 Evidence, Health, and Detector Availability

Evidence Fabric maps each source record to a NormalizedEvent with stable ID, event and ingest times, source instance, canonical subject/object/action, immutable raw pointer, and root dependency group. Forwarded copies retain their root and cannot add evidence. OCSF v1.8.0 is only a field-alignment target [10]. Health Monitor independently emits authenticated SourceCoverage intervals from heartbeat, lag, parser, and transport observations; silence is never interpreted as failure.

For $s \in S^* = \{\text{EDR}, \text{NDR}\}$,

$$m_{\epsilon, s} = 1[\epsilon \text{ contains an observed event from } s]. \quad (3)$$

Feature Extractor additionally emits a per-family detector state $a_{\epsilon, s}^{\text{det}}$. Its enum is `available`, `no_detection`, `collector_unavailable`, `parser_invalid`, or `unknown`. `no_detection` is

valid only when the collector and parser are healthy and the eligible detector stream was examined; it is distinct from missing input. Finally, $q_{\epsilon, s} = 1$ requires lineage-deduplicated detector support tied to a core entity or edge. Scope Resolver consumes (m, h, a^{det}) and exact versions; neither m nor q diagnoses health. The machine schema and valid/rejected examples are supplied with the artifact contract.

3.3 Deterministic Episode Synthesis and Revisions

For event e_i and open episode E ,

$$\ell(e_i, E) = 1[\Delta_t \leq 300s \wedge R = 1 \wedge \text{span}(E \cup \{e_i\}) \leq 3600s \wedge H = 0]. \quad (4)$$

R comprises process-causal, authenticated-session, endpoint-mapped connection, and specific-entity relations. Their precedence ρ is respectively 4, 3, 2, 1; n_R is the number of distinct satisfied relation classes. $H = 1$ when all support comes from an entity with at least 25 distinct relations in the preceding 600 s. Eligible candidates use the sole order $(-\rho, -n_R, \Delta_t, t_{\text{start}}, \text{episode_id})$.

Zero/one candidates start/attach. Multiple candidates merge only when there are at most three parents, every parent is connected through the new event by a causal, session, or endpoint relation, and the result has at most 500 events and 200 entities; otherwise the first ordered candidate receives the event. Start and merge IDs hash the policy version and canonical inputs; membership order is $(t^{\text{event}}, t^{\text{ingest}}, \text{source_instance}, \text{event_id})$.

The watermark is maximum observed ingest time minus 120 s. An episode moves `open`→`finalized` when its end precedes the watermark. Evidence accepted within the 900 s revision horizon emits $r+1$ with a predecessor pointer; later evidence is retained in the late store and cannot change online state. After the horizon the episode becomes closed. Each transition emits only EpisodeRevision; after all other producers finish, the Assembler emits the corresponding AuditRecord.

Algorithm 1: label-blind update. Validate and root-deduplicate the event; retrieve inclusive temporal candidates; compute R, H and hard guards; sort by the fixed tuple; execute start/attach/merge; advance watermark and state; then emit one immutable EpisodeRevision. Replay asserts equality of IDs, membership, links, states, features, outputs, evidence references, and canonical digests. Figure 2 illustrates the fixed transition points.

3.4 Compact Features and Stateless Raw Score

The primary scorer input is

$$x_\epsilon = [D_\epsilon, C_\epsilon]. \quad (5)$$

For eligible detector value r_i , a training-benign empirical CDF yields $d_i = \widehat{F}_{s_i}(r_i)$. D is the mean of the three greatest d_i values. With no eligible value, $D = 0$ is only a placeholder and a^{det} records the reason; any `collector_unavailable`, `parser_invalid`, or `unknown` state is out-of-support. Corroboration is fixed to

$$C_\epsilon = q_{\epsilon, \text{EDR}} q_{\epsilon, \text{NDR}}, \quad K = 2. \quad (6)$$

Identity is only an offline M5-ID annotation and cannot alter candidates or features.

Table 1: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent	Evidence Fabric	Episode Synthesis	label, split, scenario, or health fields
SourceCoverage	Health Monitor	Scope Resolver; Sufficiency Evaluator	infer outage from event silence
EpisodeRevision	Episode Synthesis	Feature Extractor; AuditRecord Assembler	scorer-side membership change
FeatureVector $[D, C, a^{det}]$	Feature Extractor	Scorer; Scope/Sufficiency; Assembler	label or inferred-health input
Raw logit z	Stateless Scorer	Calibrator; Assembler	probability wording
CalibrationStatus κ	Scope Resolver	Calibrator; Sufficiency; Assembler	test-selected status
Nullable P	Calibrator	Assembler; downstream review	modification by U ; numeric unsupported output
Structured U	Sufficiency Evaluator	Assembler; downstream review	scalarization or recomputation of P
AuditRecord	AuditRecord Assembler	Append-only store; replay; evaluation	update/delete or second producer

TrustEpisode online and offline boundaries

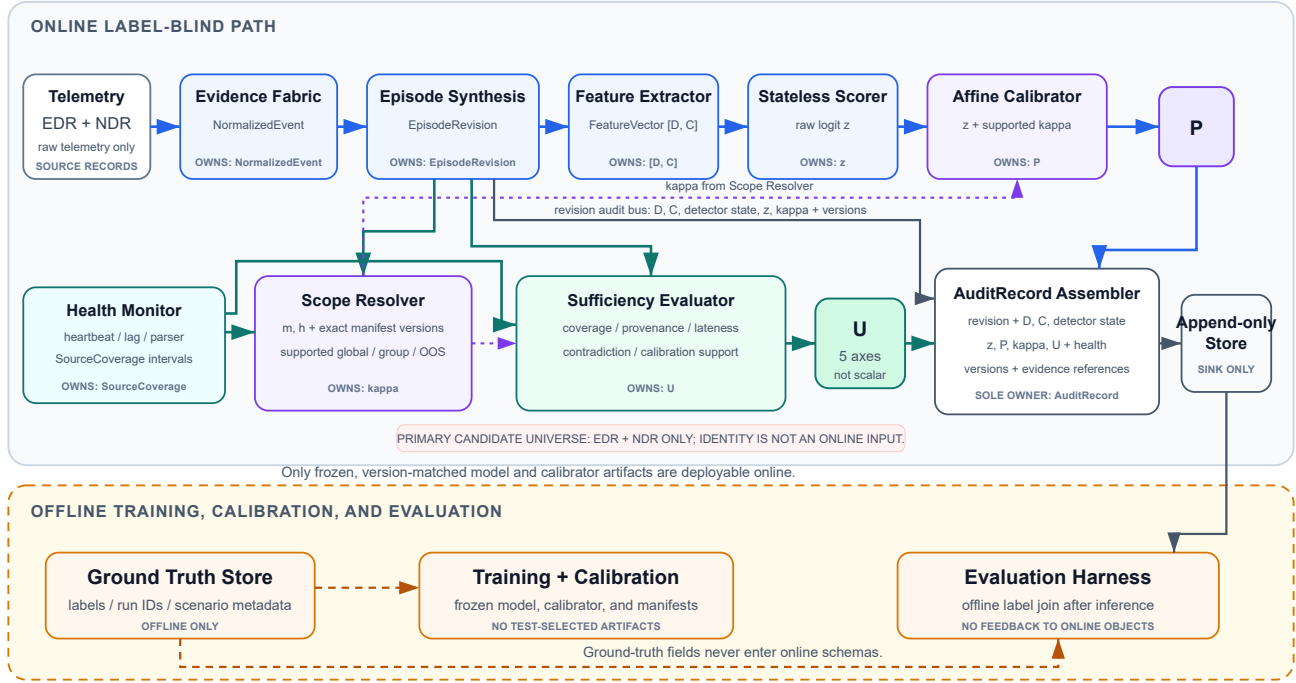


Figure 1: Label-blind online and offline boundaries. The primary candidate path contains EDR/NDR only. Every arrow enters a declared producer; the AuditRecord Assembler owns the record and the append-only store is only a sink. Identity is not an online input, and U has no numerical path into the scorer or calibrator.

The stateless scorer is

$$z_\epsilon = b + w_D D_\epsilon + w_C C_\epsilon, \quad w_D, w_C \geq 0. \quad (7)$$

Training minimizes unweighted candidate-level NLL on the training partition. Development selects the contract; then final train-only CDFs and coefficients freeze. Health, m , a^{det} , κ , U , labels, and outage status never enter x_ϵ .

3.5 Manifest-Scoped Affine Calibration

The global calibrator uniquely minimizes natural-prevalence calibration NLL plus $10^{-6}(a^2 + b_{cal}^2)$ subject to $a \geq 10^{-6}$. Thus

$$P_\epsilon = \begin{cases} \sigma(az_\epsilon + b_{cal}) & \kappa = \kappa_G, \\ \sigma(a_g z_\epsilon + b_g) & \kappa = \kappa_g, \\ \text{null} & \kappa = \kappa_\emptyset, \end{cases} \quad (8)$$

following positive-slope score calibration [4, 11]. The primary manifest supports only exact EDR+NDR mask 11, non-down/non-unknown health, valid parsers, detector states available or no_detection, and exact pipeline versions. Any mismatch is out-of-support.

Conditional groups use deterministic five-fold run-level cross-fitting (seed 20260710), never an alternative holdout. A group requires at least 50 positive revisions, 50 negatives, and 10 runs; mean held-out improvements in both Brier and NLL must be at least 0.01, every fold must be nonnegative, and both paired run-bootstrap lower 95% bounds must exceed zero. Undefined or tied gates fall back to global. Exact support plus an enabled group gives supported_group; exact support otherwise gives supported_global; mismatch gives out_of_support, null P , and decision_eligible=false. The locked test cannot change this resolver.

Deterministic episode synthesis and immutable revision lifecycle

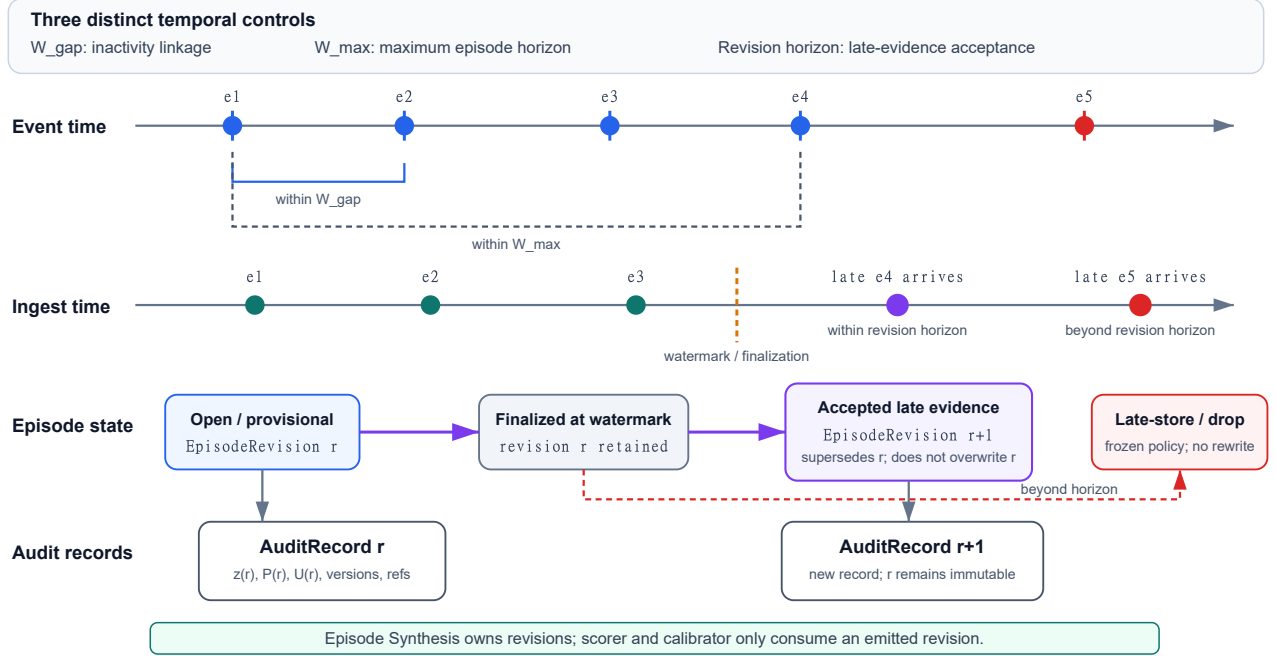


Figure 2: Deterministic lifecycle. Event time controls linking, ingest time controls the 120 s watermark, accepted evidence within 900 s creates a successor revision, and evidence beyond the horizon cannot rewrite history.

3.6 Structured Sufficiency and Audit Assembly

$$U_{\epsilon} = (u_{\text{coverage}}, u_{\text{provenance}}, u_{\text{lateness}}, u_{\text{contradiction}}, u_{\text{calibration}}). \quad (9)$$

U is never scalarized and cannot gate, modify, or relabel P . AuditRecord Assembler waits for one EpisodeRevision, FeatureVector, z , κ , P , U , and health snapshot with matching $(\text{episode_id}, \text{revision})$ and versions. Missing, duplicate, or mismatched inputs fail closed. It serializes required decision_eligible, ordered references, and canonical SHA-256, then inserts exactly once; the store only grants insert/select. Runtime host, PID, wall-clock, and duration form a non-canonical envelope excluded from the digest.

3.7 Boundary and Reproducibility Contract

The method excludes CTI, RAG/agents, impact/priority, automated response, Identity as an online source, and OT/ICS generalization. Machine schemas, thresholds, split policy, perturbations, scenario cards, and RT1–RT7 definitions are versioned in the supplied artifact bundle; the expanded appendix carries the algorithms and truth tables. The contribution is limited to deterministic EDR/NDR episode construction, manifest-scoped probability, structured sufficiency, and replayable append-only provenance.

4 EVALUATION

The evaluation is a frozen protocol, not a report of completed experiments. Populations, splits, manifests, thresholds, gates, interventions, matching, metrics, resampling, and empty RT1–RT7

schemas are fixed before locked-test access. Every unexecuted cell is pending; no synthetic value, curve, or interval is reported.

4.1 Research Questions and Populations

RQ1 – formation. Does the label-blind builder increase malicious-action coverage over time-only and time-plus-entity comparators while reducing fragmentation and over-merge? **RQ2 – ranking/calibration.** Does $D + C$ improve AUPRC and fixed-budget recall over severity and D -only comparators, and does independent affine calibration improve Brier, NLL, and reliability? **RQ3 – partial telemetry.** Under paired replay and separate physical outages, how do formation, z , support, U , revisions, and eligibility change? **RQ4 – replay/cost.** Are canonical outputs byte identical, and what throughput, latency, memory, storage, and replay time are measured?

The populations are disjoint views of the same registered runs. C_{form} contains one terminal revision per surviving episode lineage at $T_{\text{form}} = t_{\text{run_end}} + L_{\text{wm}} + W_{\text{rev}} = t_{\text{run_end}} + 1020$ s; merged parents are removed. C_{dec} contains the first finalized, decision-eligible revision per terminal episode and is the sole primary RQ2 population. C_{stream} contains every AuditRecord keyed by $(\text{episode_id}, \text{revision})$ and is used only for RQ3/RQ4 transition and cost analysis. Thus late revisions cannot silently duplicate RQ1/RQ2 candidates.

Table 2: Deterministic U rules; rows list highest-to-lowest precedence.

Axis	Precedence	First matching predicate
coverage	unknown > unavailable > degraded > complete	missing/conflicting required interval; else any down/not-collected/parser-invalid; else any degraded; else all expected healthy
provenance	missing > partial > complete	any scored/link/membership root unresolved; else optional unscored reference absent; else every required pointer and root resolves
lateness	horizon_exceeded > late_accepted > re-visible > stable	relevant evidence beyond 900 s; else successor caused by accepted late evidence; else open/finalized inside horizon; else closed
contradiction	present > unevaluable > none	any frozen online rule true; else any applicable rule lacks required online input; else every applicable rule false
calibration	out_of_support; supported_group; supported_global	exact serialization of Scope Resolver κ

4.2 Controlled Laboratory and Frozen Split

Formal claims use only the controlled Caldera v5.3.0 laboratory [1]; public provenance datasets are related-work context, not an evaluation tier. The lab contains isolated Windows/Linux targets, a benign service, EDR/NDR collectors, Health Monitor, the production-like online pipeline, a separate Ground Truth Store, and an offline Evaluation Harness. The release pin is annotated tag object dd22d543... peeled to commit b24f6e7a...; every formal run additionally records target, collector, configuration, topology, parser, and schema digests. Missing manifest fields fail closed into RT1 only.

Identity is never an online source. M5 is an EDR/NDR endpoint card; M5-ID is a separately labeled post-inference annotation admitted only after its timestamp/provenance/health/support pilot passes. It cannot alter S^* , $K = 2$, candidates, features, manifests, or primary RQs.

4.3 Ground Truth, Matching, and Labels

Allowlist schemas reject operation_id, ability_id, adversary/technique labels, attack markers/windows, scenario class, malicious label, and split assignment from every online object [6]. Offline matching first requires temporal IoU ≥ 0.10 , entity Jaccard ≥ 0.50 , and one exact harmless marker or registered action. Edge weight is

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2IoU_{time}; \quad w \geq \theta_{match} = 0.50. \quad (10)$$

Maximum-weight bipartite assignment operates within one run; ties use entity overlap, absolute time difference, then lexical IDs.

A revision is positive when malicious-action coverage $\geq \theta_{mal} = 0.50$ and unrelated benign contamination $< \theta_{contam} = 0.25$; mixed when both thresholds are met; negative when malicious coverage is zero and activity is attributable to a registered benign/background source; all residual, missing, or conflicting cases are ambiguous. Labels are joined only after AuditRecord insertion. Sensitivity varies $\theta_{mal} \in \{.25, .50, .75\}$, $\theta_{contam} \in \{.10, .25, .50\}$, and $\theta_{match} \in \{.40, .50, .60\}$ without altering the primary registry. Figure 3 shows the enforced join boundary.

4.4 Scenario and Benign Cards

B1–B8 respectively cover inventory, patching, file synchronization, vulnerability scanning, helpdesk administration, log collection, configuration management, and declared unscheduled maintenance, with 10 pure-benign repetitions each. Every card fixes preconditions, harmless markers, expected observations, cleanup, and card-specific failure. Collection failure remains in RT1/failure logs but leaves RT2–RT7; action failure with healthy

collectors enters only the sensitivity cohort; cleanup failure blocks image reuse. The supplied scenario-card contract contains the ordered M1–M7, M5-ID, and B1–B8 definitions.

4.5 Calibration Protocol and Comparators

Training fits the raw scorer; development freezes the contract; calibration fits the global mapping and executes the Section 3.5 five-fold group gate; locked test selects nothing. Supported results report counts, prevalence, Brier/NLL, ten equal-width reliability bins with counts, and run-cluster intervals. Unsupported records report coverage and status but receive no proper score. The global positive-slope mapping must preserve AUPRC/AUROC on C_{dec} or trigger an implementation audit.

4.6 Replay Perturbations and Physical Outage

Every M1–M7 base run has deterministic children using an HMAC-SHA256 child seed over base ID, intervention, and level with key 20260710. Source masks are EDR-only/NDR-only; independent drop rates are 5/10/25%; burst drops are 30/120/300 s; fixed ingest delays and reorder buffers are 30/120/600 s; event-time skew is $-120/-30/+30/+120$ s; duplicate rates are 5/10/25%; and RP0 is unchanged. Replay never changes SourceCoverage. A physical EDR or NDR outage follows 120 s steady state, lasts 120 or 600 s, and then restores the collector. Recovery requires two healthy heartbeats, valid parser, backlog ≤ 120 s, and no new revision for 900 s. Pair descendants inherit split and cluster. Figure 4 summarizes the non-pooled designs.

4.7 Metrics and Run-Level Uncertainty

For run r , let G_r be its ground-truth action groups and n_g the number of distinct terminal predicted episodes assigned to group g . RQ1 uses run-macro means of action coverage, $|G_r|^{-1} \sum_g \max(0, n_g - 1)$ fragmentation, and over-merge: the fraction of terminal predicted episodes assigned to more than one distinct group in the same run. This cannot degenerate through mutually exclusive run IDs. Contamination and one-to-one episode F1 are secondary. Matching uses Equation 10.

RQ2 uses C_{dec} only: AUPRC and recall among the top $\max(1, \lceil 0.05|C_{dec,r}| \rceil)$ per run are primary ranking metrics; Brier, NLL, and ten-bin reliability use supported records only. AUROC and precision@ k are secondary [12]. RQ3 reports paired run-macro changes and all U transitions; RT5 reports outage detection/recovery. RQ4 checks byte equality and measured cost.

All primary summaries are macro-averaged over runs; pooled micro results are supplemental. Intervals use 2,000 percentile bootstrap resamples of complete run clusters with seed 20260710 [3];

Table 3: Frozen campaign split and descendant inheritance.

Item	Unique rule	Fail-closed invariant
Independent unit	base campaign, stratified by scenario family and benign mode	no event/episode random split
Order	ascending SHA-256 of 20260710:base_campaign_id per stratum	freeze IDs before label release
20-run cycle	ranks 0–9 train; 10–13 development; 14–16 calibration; 17–19 locked test	target 50/20/15/15 percent
Descendants	replay/outage children inherit partition and bootstrap cluster	any mismatch invalidates the complete cluster
Registry	run, base/parent/pair IDs, scenario, benign mode, seed, partition, cluster, role, manifest digest	missing field excludes RT2–RT7

Ground Truth Firewall and offline label join

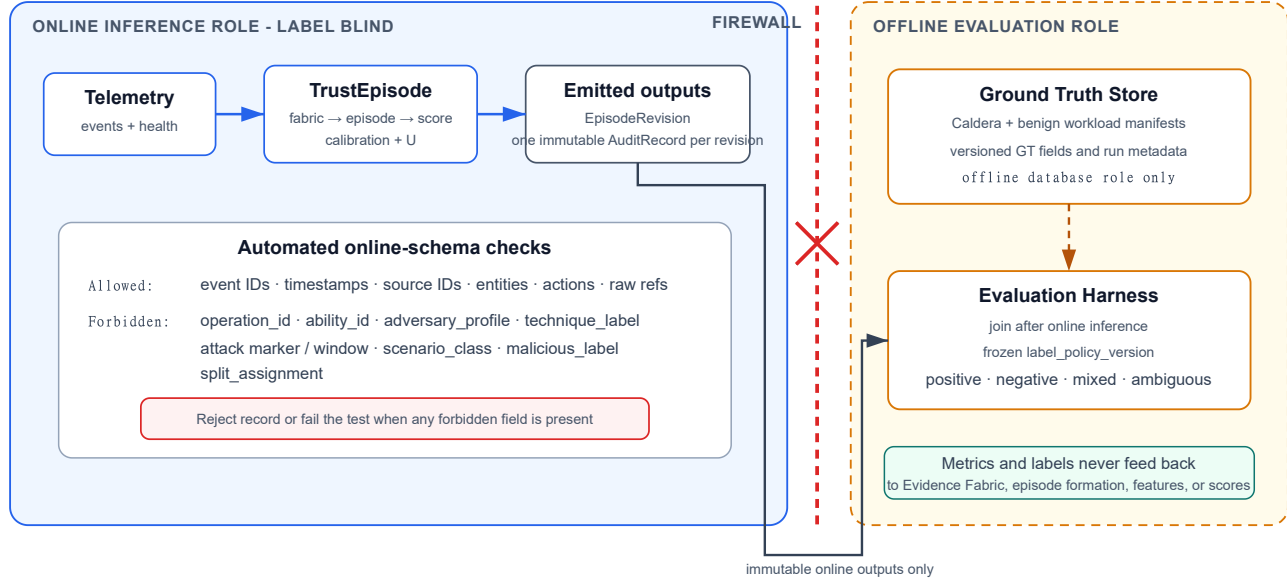


Figure 3: Ground Truth Firewall. Schema tests reject evaluation-only fields before inference; labels join immutable outputs only inside the offline Evaluation Harness.

Table 4: Frozen malicious families; every card has 10 successful repetitions in attack-only and concurrent-benign modes.

ID	Ordered family	Stressor	Primary evidence
M1	Windows discovery → share → remote marker	lateral chain	EDR lineage + NDR connection
M2	script → marker transfer → temporary task	weak-to-strong	EDR lineage + NDR transfer
M3	Linux discovery → test SSH → file marker	cross-platform	Linux EDR + NDR session
M4	low-rate connections → local marker	sparse sequence	NDR cadence + EDR action
M5	test-account endpoint action → endpoint marker	Identity-independent primary card	EDR/NDR only
M6	dual-use admin tool → remote marker	tool-name shortcut	contextual EDR/NDR
M7	discovery → fixed pauses → remote marker	300/3600 s boundaries	fragmentation/stabilization

Table 5: Registered comparisons; each ID has one implementation and one result location.

ID	Scope	Single implementation	Purpose / registry
EB1	formation	non-overlapping 300 s event-time windows	time-only / RT2
EB2	formation	EB1 plus one specific shared entity	time+entity / RT2
EB3	formation	proposed guarded temporal-causal builder	reference / RT2
EB3-H	formation	EB3 with hub suppression disabled	component ablation / RT2
SB1	ranking	maximum normalized source-native severity	external rank / RT3a
SB2	ranking	constrained D -only logit	corroboration ablation / RT3a
SB3	ranking	proposed $D + C$ logit	raw rank / RT3a
SB4	score	uncalibrated sigmoid score $\sigma(z)$	non-probability comparator / RT3b
SB5	calibration	global affine mapping	primary / RT3b
SB6	calibration	enabled registered group mapping	conditional / RT3b
SB7	replay	unavailable detector forced to zero; support ignored	anti-pattern / RT4
OB1	outage	available-source denominator; < 2 healthy gives no corroboration	outage only / RT5

replay/outage resample base-run pairs. Undefined class-dependent

resamples are counted and omitted, never zeroed. Relative effects are supplemental.

Paired partial-telemetry perturbations and separate physical-outage validation

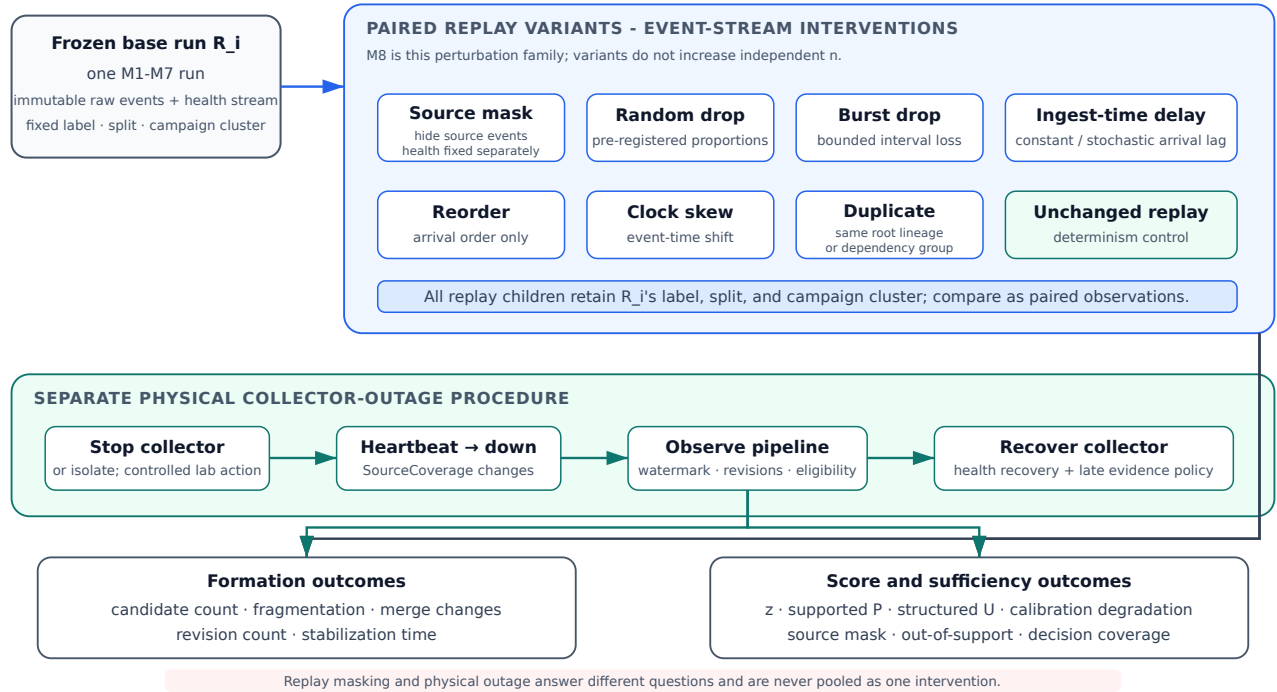


Figure 4: Paired replay changes events while health stays fixed; physical outage changes authenticated health and is evaluated separately.

4.8 RT1–RT7 Reporting Contract

Planned reliability and revision plots remain absent until locked outputs exist. The supplied machine registry contains the same RT keys, rows, columns, pending, and N/A rules; moving shells does not change the preregistration.

4.9 Limitations and Safety

The controlled lab cannot establish field generalization or every evasion/base rate. Calibration applies only to exact supported manifests. Identity remains offline. Formal runs use isolated hosts, test accounts, harmless markers, cleanup gates, and recorded failures. The evaluation does not study CTI, agents, impact, priority, OT/ICS, or autonomous response, and reports no empirical result until the locked pipeline executes.

Table 6: Frozen result registries. Every unexecuted field serializes as pending; only structurally inapplicable fields use N/A.

Registry	Rows	Frozen columns	State
RT1	partition $\times$ population	runs; terminal/decision/stream counts; positive/negative/mixed/ambiguous; prevalence	pending
RT2	EB1, EB2, EB3, EB3-H	action coverage; fragmentation; over-merge; contamination; episode F1	pending
RT3a	SB1, SB2, SB3	AUPRC; recall@budget; AUROC; precision@ k	pending
RT3b	SB4, SB5, SB6	Brier; NLL; reliability bins; supported coverage; OOS rate	pending
RT4	RP0–RP7 $\times$ level	formation/z/Brier/NLL changes; coverage; U transitions; revisions; stabilization	pending
RT5	source $\times$ duration $\times$ method/OB1	health detection; eligibility; OOS; recovery; stable revision time	pending
RT6	worker count	hash equality; throughput; p50/p95/p99; memory; bytes/event/revision; replay time	pending
RT7	error category	count; run/candidate fraction; representative revision; adjudication	pending

5 CONCLUSION

TRUSTEPISODE defines a narrow, replayable contract for EDR/NDR episode formation and candidate-conditioned maliciousness calibration. Episode Synthesis alone owns immutable membership revisions; the scorer, support resolver, calibrator, sufficiency evaluator, and AuditRecord Assembler have disjoint outputs; and missing or incompatible telemetry yields an explicit out-of-support state rather than a fabricated probability. The accompanying protocol fixes implementation and evaluation decisions before locked-test access. Empirical results remain pending, so the present contribution is the deterministic method and preregistered evaluation contract, not a claim of measured superiority or deployment generalization.

A ONLINE DATA CONTRACTS AND OWNERSHIP

A.1 Canonical Online Schemas

All online objects are versioned, immutable after emission, and validated against an allowlist before storage. Table 7 defines the minimum contract. Optional fields may be absent but cannot change the meaning of a required field.

NormalizedEvent is produced only by Evidence Fabric; SourceCoverage only by Health Monitor; EpisodeRevision only by Episode Synthesis; FeatureVector only by Feature Extractor; z only by Stateless Scorer; κ only by Scope Resolver; P only by Calibrator; U only by Sufficiency Evaluator; and AuditRecord only by AuditRecord Assembler. The store is only a sink: it grants INSERT and SELECT but denies UPDATE and DELETE for every emitted revision and record. A correction is represented by a new revision and record that references its predecessor.

A.2 Types, Time, Pointers, and Enums

Identifiers are non-empty UTF-8 strings scoped by their producer version. Event and ingest times are UTC RFC 3339 timestamps with microsecond precision. Raw pointers contain an immutable object identifier, content digest, byte range or record locator, and storage schema version; they never embed mutable file paths as identity. Health is one of {healthy, degraded, down, unknown, not_applicable}. Every evidence reference includes its root dependency group so that forwarded or derived copies remain one observation.

A.3 Canonical Serialization and Digest

The deterministic payload uses UTF-8 RFC 8785 JSON Canonicalization Scheme bytes, including lexicographically ordered object keys and shortest round-trip decimal encodings. Arrays use the schema’s explicit domain order. NaN and infinities are forbidden. When $\kappa = \kappa_\emptyset$, the serialized probability field is the JSON literal null; it is never omitted, zeroed, or replaced by an estimate. The canonical digest is SHA-256 over these bytes. Wall-clock execution time, host name, process ID, memory address, and machine-specific runtime counters are stored as a separate non-canonical envelope and cannot affect the digest.

The supplied schema sets additionalProperties to false for every object. It lists all required and nested types, fixes enum

values, and requires decision_eligible. It is versioned as trustepisode_online.v1.schema.json. The AuditRecord digest include set is every required field except canonical_digest; the runtime envelope is also excluded. Included examples validate a supported record and a null-probability unsupported record. A third example containing malicious_label is required to fail schema validation.

A.4 Firewall Examples

A valid minimal record contains only online-observable fields:

```
event_id=edr:0042;
event_time=2026-07-10T04:00:00.000000Z;
ingest_time=2026-07-10T04:00:01.120000Z;
source=EDR; subject=proc:p7; object=file:f9;
action=write; raw_ref=sha256:...;
dependency_group=root:0042; parser_version=p3.
```

The firewall must reject the same record if it includes, for example, ability_id=T1059-001 or malicious_label=1. Rejection fails the run-level schema test, records the offending producer, and prevents the object from entering Episode Synthesis. Merely ignoring a forbidden field downstream is insufficient.

B EPISODE SYNTHESIS AND REVISION ALGORITHM

B.1 Inclusive Eligibility and Allowed Relations

Both temporal comparisons in Equation 4 are inclusive with $W_{\text{gap}} = 300$ s and $W_{\text{max}} = 3600$ s: $\Delta_t = W_{\text{gap}}$ and $\text{span}(E \cup \{e_i\}) = W_{\text{max}}$ remain eligible. Values larger than either bound are ineligible. The relation predicate R is true when at least one of the following frozen, namespace-aware relations holds:

- (1) process parent-child or process-to-file causal continuity;
- (2) authenticated local/remote session continuity;
- (3) a network connection with a specific source/destination endpoint mapping;
- (4) the same non-generic host, user, process, file, service, or session entity.

The hub predicate H is true when all apparent support is attributable only to an entity with at least 25 distinct relations in the preceding 600 s. Dependency-linked duplicates are collapsed before R and H are evaluated.

B.2 Total Order and Branch Semantics

Candidate episodes are ordered by the tuple

$$(-\rho, -n_R, \Delta_t, t_{\text{start}}, \text{episode_id}), \quad (11)$$

where ρ is the strongest relation precedence, n_R is the number of distinct allowed relation classes, Δ_t is the smallest event-time gap, t_{start} is episode start time, and the final key is lexical. Relation precedence is process-causal, authenticated session, endpoint-mapped connection, then specific-entity equality. This tuple is the only tie-break order.

With zero eligible candidates, the builder starts an episode whose ID is the SHA-256 digest of the formation-policy version, first event ID, and canonical namespace. With one candidate, it attaches the event. With multiple candidates, it merges only

Table 7: Canonical online schema registry.

Object	Key	Required content	Permitted optional content
NormalizedEvent	event_id	UTC event/ingest times; source family/instance; canonical subject, object, action; raw pointer; dependency group; schema/parser versions	source-native attributes on an explicit allowlist
SourceCoverage	source instance plus interval start	interval end; expected flag; health enum; observed lag; reason code; monitor/policy versions	authenticated diagnostic reference
EpisodeRevision	episode ID plus revision number	ordered event IDs; accepted link reasons; parent/merge lineage; event-time bounds; watermark; formation-policy version; membership digest	late-evidence reason and superseded revision ID
FeatureVector	episode ID plus revision number	D, C ; per-family detector state and missing reason; feature version; normalizer hashes; ordered input references	none
AuditRecord	episode ID plus revision number	EpisodeRevision digest; $D, C, a^{det}, z, P, \kappa, U$; decision_eligible; versions; health/mask snapshots; ordered references; canonical digest	non-canonical runtime envelope

Frozen label-blind attach-merge-start decision procedure

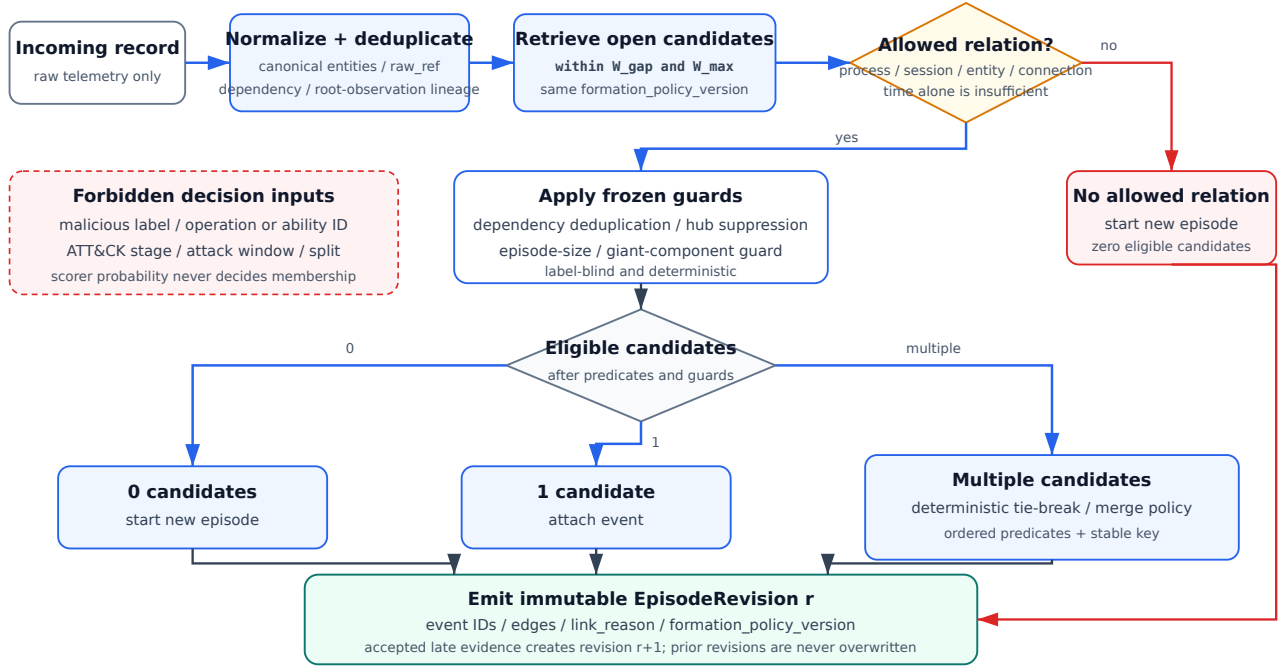


Figure 5: Frozen label-blind attach-merge-start procedure. Candidate retrieval, relation checks, guards, and total ordering execute before Episode Synthesis emits a revision; labels and scorer outputs cannot influence membership.

when there are at most three parents, every parent is connected through the new event by a process-causal, authenticated-session, or endpoint relation, and the result has at most 500 events and 200 entities; otherwise it attaches to the first ordered candidate. A merge ID is the digest of the literal merge, the sorted parent IDs, and the formation-policy version. Parent IDs and their last revision numbers are retained as merge lineage.

Within a revision, events are ordered by

$$(t^{event}, t^{ingest}, \text{source_instance}, \text{event_id}). \quad (12)$$

The revision digest covers the canonical episode ID, revision number, ordered membership, ordered link reasons, lineage, and formation-policy version. These rules eliminate dependence on thread scheduling or database return order.

B.3 Watermark and Late Evidence

The watermark is maximum observed ingest time minus $L_{wm} = 120$ s. An episode moves open→finalized when its event-time end precedes the watermark. After finalization, accepted evidence inside $W_{rev} = 900$ s emits $r + 1$ with a predecessor pointer. At the horizon it becomes closed; later evidence is retained in the late store, emits no online revision, and cannot alter history. Event time is never overwritten to make late evidence appear timely.

B.4 Deterministic Procedure and Replay Test

- (1) Validate the NormalizedEvent schema and deduplicate by root dependency group.
- (2) Retrieve open episodes whose inclusive time bounds can admit the event.
- (3) Compute allowed relation classes and reject hub-only support.

- (4) Apply the 3-parent, 500-event, 200-entity, and strong-relation merge guards.
- (5) Sort candidates exactly by Equation 11.
- (6) Execute the zero/one/multiple branch and construct canonical lineage.
- (7) Order membership, compute the revision digest, and emit EpisodeRevision.
- (8) Compute $D, C, a^{det}, z, \kappa, P, U$; the AuditRecord Assembler validates matching keys and inserts one record.
- (9) Advance the watermark; process accepted late evidence only as a new revision.

The replay test runs the same normalized input twice under different worker counts and database insertion orders. It compares episode/revision IDs, ordered memberships, link reasons, lineage, features, scores, statuses, U , evidence references, and canonical digests. Any mismatch is a correctness failure, not measurement noise.

C FEATURE, SUFFICIENCY, AND PARAMETER REGISTRY

C.1 Feature and Supporting-Evidence Contract

For source family s , the empirical CDF artifact $\hat{F}_s^{\text{train, benign}}$ is fit on training-benign detector values after source parser validation. Ties use the right-continuous empirical CDF. The artifact records sample count, fitting window, source contract, parser version, and content hash. Episode-level D_ϵ is the arithmetic mean of the three greatest eligible transformed values. The fixed top-three constant is not learned.

$q_{\epsilon, s} = 1$ requires an upstream detection record with an immutable raw pointer, root dependency group, detector-contract version, and a verified relation to an episode core entity or edge. Coverage heartbeat, inventory, unscored raw presence, and another transport of the same root observation are ineligible. The primary corroboration feature remains $C = q_{\text{EDR}} q_{\text{NDR}}$ with fixed $K = 2$.

Feature Extractor emits $a_{\epsilon, s}^{det}$ as available, no_detection, collector_unavailable, parser_invalid, or unknown. If a healthy, valid detector stream was examined and had no eligible value, $a^{det} = \text{no_detection}$ and $D = 0$ is a measured placeholder; provenance can remain complete. The last three states also use $D = 0$ but are missing inputs, force out-of-support, and are never interpreted as negative evidence.

C.2 Complete Sufficiency Rules

The contradiction registry can compare online facts such as incompatible process lineage or conflicting endpoint mappings. It cannot read a Caldera operation, ability, attack window, scenario family, split, or offline label. Multiple rules combine by the deterministic precedence present over unevaluable over none. The five axes are serialized in the table order and are never converted to one number.

C.3 Parameter Taxonomy

The unconditional primary method therefore has five fitted scalar coefficients: $\{b, w_D, w_C, a, b_{\text{cal}}\}$. Empirical CDFs are frozen fitted preprocessing artifacts, not fixed constants. Conditional group coefficients exist only for groups that pass the registered gate.

C.4 Single Threshold Registry

Every threshold is defined only in Table 10; other sections refer to its symbol and owner.

Primary values are $W_{\text{gap}} = 300$ s, $W_{\text{max}} = 3600$ s, $W_{\text{rev}} = 900$ s, $L_{\text{wm}} = 120$ s, $f_{\text{hub}} = 25$ distinct relations per 600 s, and $n_{\text{component}} = (500 \text{ events}, 200 \text{ entities})$. Label/matching values are $\theta_{\text{mal}} = .50$, $\theta_{\text{contam}} = .25$, and $\theta_{\text{match}} = .50$ with minimum time IoU .10 and entity Jaccard .50. The group gate fixes $(n_{\text{pos}}^{\text{min}}, n_{\text{neg}}^{\text{min}}, n_{\text{run}}^{\text{min}}) = (50, 50, 10)$, five run folds, and minimum mean Brier/NLL improvements (.01, .01) with positive paired 95% lower bounds. Values are mirrored in thresholds.v1.json; a code/paper mismatch fails artifact validation.

D CALIBRATION MANIFEST, ELIGIBILITY, AND INVALIDATION

D.1 Manifest Contract

Support requires exact equality for every contract field that affects candidate construction, features, raw scores, or calibration meaning. The resolver never chooses a nearest source mask or silently maps a novel pipeline version to an old artifact.

D.2 Global and Conditional Group Resolver

The global mapping uniquely minimizes natural-prevalence NLL plus $10^{-6}(a^2 + b_{\text{cal}}^2)$ subject to $a \geq 10^{-6}$. Primary support requires exact mask 11, no down/unknown health, valid parsers, and detector states available or no_detection. A pre-registered group is considered only with at least 50 positives, 50 negatives, and 10 independent runs. Runs use deterministic five-fold assignment with seed 20260710; each fold evaluates a mapping fitted on the other four. A group is enabled only when mean held-out improvements in both Brier and NLL are at least .01, every fold is nonnegative, and both paired run-bootstrap lower 95% bounds exceed zero. Undefined or tied gates fall back to global. The final group coefficients are then fit on all calibration runs; the locked test never selects a group.

Resolution follows one exclusive branch:

- (1) Exact manifest match plus enabled registered group: use (a_g, b_g) and serialize supported_group.
- (2) Exact manifest match without an enabled group: use (a, b_{cal}) and serialize supported_global.
- (3) Any required manifest or supported-regime mismatch: serialize out_of_support, set $P = \text{null}$, and set decision_eligible=false.

Group-gate failure is therefore distinct from out-of-support status.

Table 8: Complete deterministic rules for U .

Axis	Value	Rule
coverage	unknown unavailable degraded complete	first: any required health interval is missing or conflicting else any expected instance is down/not collected or any parser is invalid else at least one expected instance is degraded else every expected instance is healthy and parser valid
provenance	missing partial complete	first: any scored value, accepted link, membership event, or required root is unresolved else at least one optional unscored reference is absent else every required pointer and dependency root resolves
lateness	horizon-exceeded late-accepted revisable stable	first: relevant evidence arrived beyond 900 s and was late-store only else this revision was created by accepted late evidence else episode is open or finalized inside the 900 s horizon else episode is closed
contradiction	present unevaluable none	first: any applicable online rule identifies conflicting observations else any applicable rule lacks required online input else every applicable frozen online rule is false
calibration	supported_global supported_group out_of_support	κ = supported_global κ = supported_group κ = out_of_support

Scoped probability calibration and independent evidence sufficiency

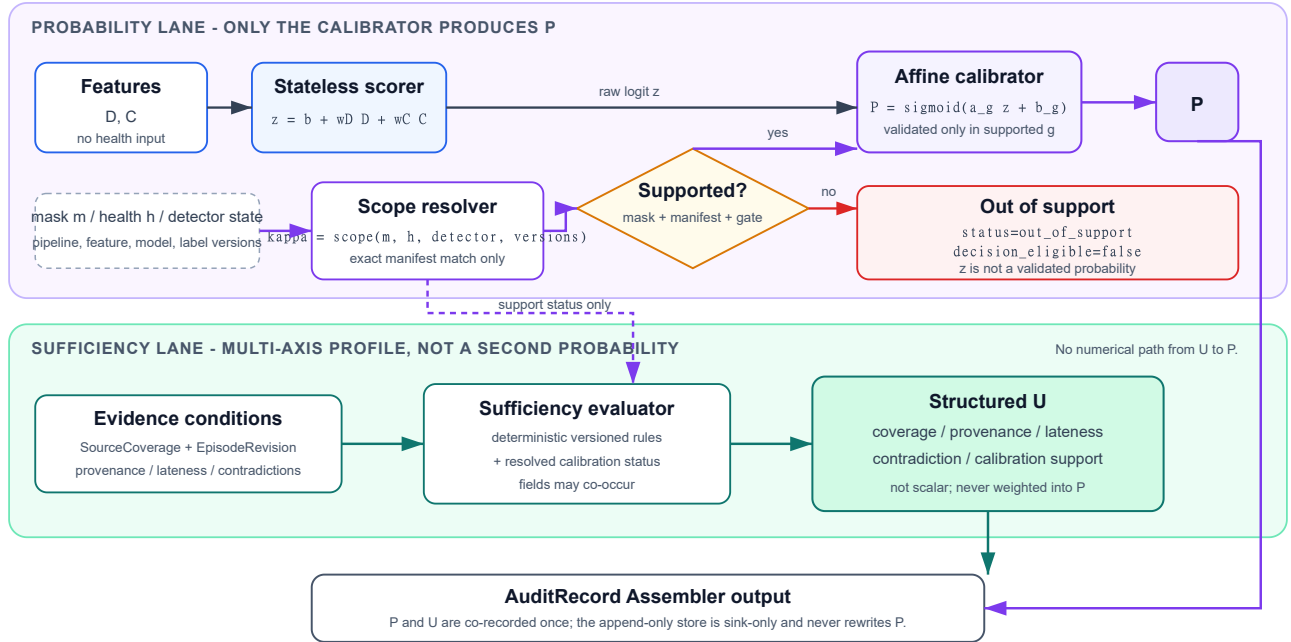


Figure 6: Probability calibration and evidence sufficiency are independent lanes. Detector state participates in support resolution, while U has no numerical path into P ; the AuditRecord Assembler emits one record to a sink-only store.

D.3 Invalidation Matrix

Prevalence is recorded because it affects interpretation and may motivate revalidation, but an unknown current deployment prevalence is never an online per-revision gate. Out-of-support records expose z only with raw-logit wording and serialize P as canonical null under Appendix A.3.

E LABORATORY, GROUND TRUTH, AND LABEL POLICY

E.1 Run and Environment Manifest

Every formal run stores: run and pair IDs; scenario and benign-workload manifest versions; Caldera v5.3.0 annotated tag object dd22d543a519afe63ca35c454bf1add5d75c45dc, peeled commit b24f6e7a99cab19cbf417009bda9b9c6c81abc31; target image SHA-256 hashes; VM and network topology versions; EDR/NDR collector images and configuration hashes; Health Monitor version; parser/schema versions; clock source and maximum observed skew; random seed; start/end times; harmless marker IDs; cleanup

Table 9: Parameter and state taxonomy.

Category	Content	Learned?	Owner	Selection/fit data	Freeze point
Scorer coefficients	(b, w_D, w_C)	yes	Stateless Scorer	training fit; development selects contract	before calibration
Global calibrator	(a, b_{cal})	yes	Calibrator	natural-prevalence calibration split	before locked test
Conditional calibrator	(a_g, b_g)	yes	Calibrator	calibration run folds after gate	before locked test
Formation policy	time, revision, hub, component, merge settings	no	Episode Synthesis	training/development	before final train-only fit
Fixed aggregation	top three; primary $K = 2$	no	Feature contract	method specification	before training
Evidence state	m, h, q, a^{det}	no	Table 1 producers	runtime observation	per revision
Output/status	κ, U	no	Scope/Sufficiency Evaluators	deterministic runtime rules	per AuditRecord
Version metadata	schema, policy, model, calibrator, manifest hashes	no	Artifact Registry	producing artifact’s split	artifact publication
Evaluation settings	split seed, perturbations, bootstrap repetitions	no	Evaluation Harness	pilot/development preregistration	before locked test

Table 10: Threshold ownership, selection, freeze, and validation registry.

Symbol	Meaning	Owner	Selection data	Freeze and validation
W_{gap}	maximum event-time inactivity gap	formation policy	development	before final train fit; EB1/EB2 sensitivity
W_{max}	maximum event-time episode span	formation policy	development	before final train fit; M7 sensitivity
W_{rev}	post-watermark revision horizon	revision policy	development/pilot	before calibration; delay replay
L_{wm}	allowed ingest lateness for watermark	revision policy	development/pilot	before calibration; delay replay
f_{hub}	generic/high-frequency entity cutoff	formation policy	training/development	before final train fit; EB3-H ablation
$n_{component}$	giant-component event/entity limit	formation policy	development	before final train fit; stress replay
θ_{mal}	minimum malicious overlap for positive label	label policy	pilot/adjudication	before partitioned labels are released
θ_{contam}	unrelated-benign overlap for mixed label	label policy	pilot/adjudication	before partitioned labels are released
n_{pos}^{min}	minimum group-calibration positives	calibration gate	pre-registered before calibration	before group fitting; internal run folds
n_{neg}^{min}	minimum group-calibration negatives	calibration gate	pre-registered before calibration	before group fitting; internal run folds
n_{run}^{min}	minimum independent calibration runs	calibration gate	pre-registered before calibration	before group fitting; internal run folds
Δ_{miscal}^{min}	minimum validated global-calibration deficiency	calibration gate	pre-registered before calibration	internal run folds only
θ_{match}	minimum action/entity overlap for episode matching	evaluation policy	development	before locked test; matching sensitivity

Table 11: Required calibrator-manifest fields.

Field group	Required values
Candidate contract	formation-policy version; candidate-selection version; label-policy version
Feature contract	feature version; EDR/NDR detector-contract versions; empirical-CDF hashes; detector-state and missing-value contract
Model contract	raw-model version and digest; coefficient digest; training split manifest
Calibration contract	calibrator version; global coefficients; eligible group coefficients; calibration split manifest
Runtime support	exact source mask 11; allowed health/detector states; schema/parser versions; deployment/pipeline version
Evidence and audit	calibration counts, prevalence as metadata, gate evidence, reliability-bin specification, manifest digest

Table 12: Calibrator invalidation and revalidation rules.

Change	Immediate status	Required action
Episode policy or candidate-selection policy	invalid	rebuild candidates; refit scorer and calibrator
Feature semantics, normalizer, or missing rule	invalid	refit scorer and calibrator
Upstream detector contract or parser semantics	out of support	revalidate features; refit if distribution/meaning changes
Raw-model coefficients or training population	invalid	refit calibrator
Label policy	invalid for supervised claim	regenerate labels; repeat fit and evaluation
Supported source mask or health regime	out of support	collect calibration support and register a new manifest
Schema-only additive field excluded from features	pending compatibility check	prove canonical equivalence before manifest amendment
Candidate prevalence drift	revalidation signal only	audit reliability; do not automatically change episode-level output

status; and any collection incident. Formal evaluation has no public-data tier: provenance datasets remain related-work context because they do not implement the frozen EDR/NDR collection contract. Formal results group by this run ID, not by Caldera ability or individual episode. The complete machine contract

is artifacts/contracts/lab_manifest.v1.json; a missing required field fails closed.

Calibration validity, group eligibility, and artifact invalidation

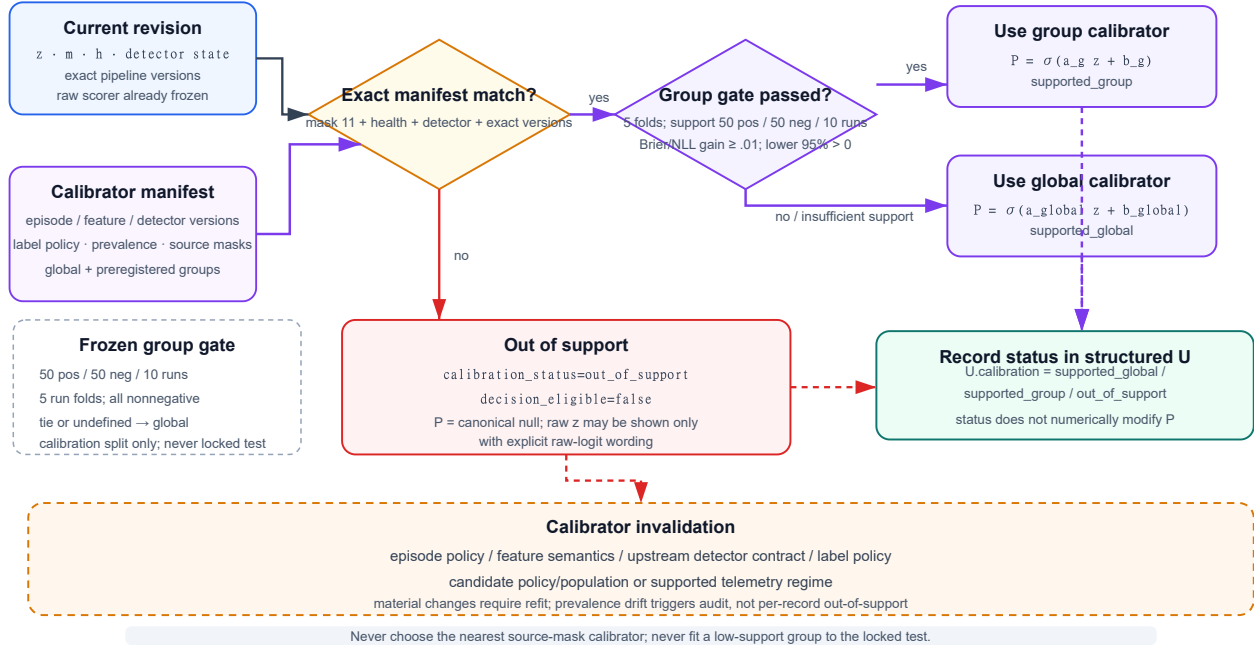


Figure 7: Exclusive calibration resolution and invalidation. Exact source, health, detector, and version support is required; the five-fold group gate can only select between a registered group and global fallback, never manufacture support.

Table 13: Malicious scenario-card registry. Actions use only isolated targets and harmless per-run markers.

ID	Precondition and ordered actions	Marker and expected observations	Cleanup and card-specific failure
M1	two Windows targets and test share; discovery → share access → test-account remote execution	unique process/file marker; EDR lineage plus NDR remote connection	remove marker/session/service; abort if either primary collector fails
M2	Windows target; script execution → staged marker transfer → temporary scheduled-task persistence	script/process/file marker; EDR lineage and NDR transfer	delete task and marker; record inability to remove persistence as cleanup failure
M3	Linux targets; discovery → test-account SSH → marker file action	shell/file marker; Linux EDR lineage and NDR SSH flow	terminate shell, revoke key, remove marker; abort on namespace mismatch
M4	instrumented service; low-rate periodic connections → local marker action	seeded connection cadence plus EDR local action	stop client and remove marker; fail card if cadence manifest is not replayed
M5	test-account endpoint action → endpoint marker	EDR/NDR primary evidence independent of Identity	revoke credential and marker; fail if either primary collector is unhealthy
M6	dual-use administration tool also present in benign cards; contextual misuse → remote marker action	identical tool name in both classes; sequence/context plus EDR/NDR distinguish use	remove marker/session; preserve unsuccessful execution as declared failure
M7	extended schedule; discovery → frozen pauses → remote action → marker	weak observations across $W_{\text{gap}}/W_{\text{max}}$ boundaries	terminate sessions/markers; report any schedule-bound violation

Table 14: Benign and hard-negative workload-card registry.

ID	Precondition and ordered actions	Expected observations	Cleanup and card-specific failure
B1	approved PowerShell/bash inventory over registered targets	process, query, and normal connection evidence	remove output; fail if command leaves target state changed
B2	signed test patch deployment and verification	installer lineage, package writes, repository flow	uninstall/revert snapshot; record reboot or roll-back failure
B3	seeded file-share synchronization of benign fixtures	burst file operations and SMB/NFS flows	remove synchronized fixtures; verify checksums
B4	authorized vulnerability scan against isolated service	high fan-out NDR flows and expected service logs	stop scanner; fail if target leaves healthy state
B5	helpdesk test-account remote administration	same remote/admin tools used in M1/M6 without malicious marker	terminate session and revoke temporary grant
B6	scheduled log collection and forwarding	repeated reads, compression, and collector-bound network flow	remove bundle; verify no collector backlog
B7	configuration-management convergence to a known benign state	remote orchestration, file/service updates, expected restart	restore baseline; fail on manifest drift
B8	unscheduled but declared maintenance with a frozen work order	realistic administrative sequence with healthy EDR/NDR	verify service state; retain timing as hard-negative context

E.2 Scenario Cards and Failure Rules

Every card records its version, target images, benign mode, repetition ID, and seed. Each malicious family has exactly 10

successful attack-only repetitions and 10 successful concurrent-benign repetitions; B1–B8 each have exactly 10 successful

Controlled Caldera laboratory and production-like telemetry path

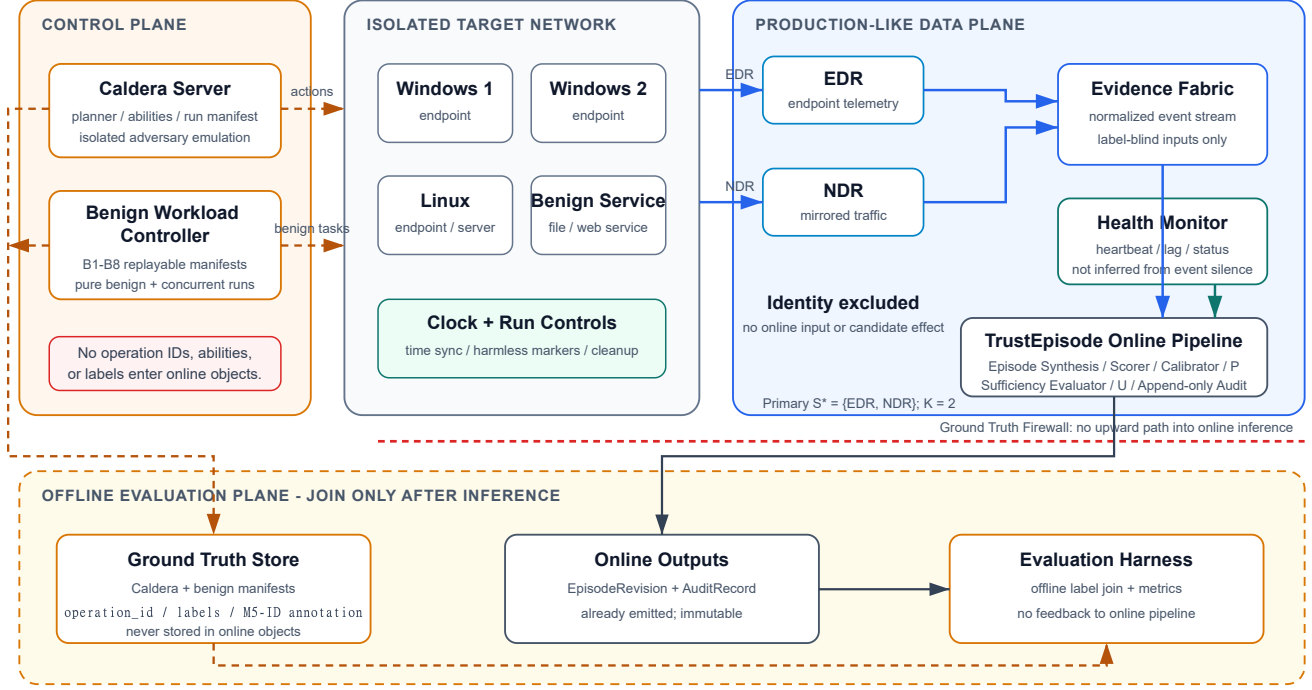


Figure 8: Controlled laboratory topology. EDR and NDR are the only online telemetry sources; Identity is excluded from the candidate path and M5-ID can join only in offline evaluation.

pure-benign repetitions. The machine-readable ordered actions, markers, observations, cleanup, and failure rules are in `artifacts/contracts/scenario_cards.v1.json`. A collection failure excludes the run from primary metrics but remains in RT1 and the failure log. A failed scenario action with healthy primary collectors remains in the registered sensitivity view but not the successful-run cohort. Cleanup failure blocks the next run on that image. No failed or aborted run is silently reused under the same ID.

E.3 Identity Pilot Gate

Identity remains conditional under the separately registered M5-ID card. Before any Identity-side analysis, pilot runs must demonstrate complete timestamp alignment, stable benign and malicious capture, independent source health, raw provenance pointers, and sufficient run-level support. Identity is always excluded from `NormalizedEvent`, `Episode Synthesis`, candidate IDs and membership, D, C, z , every calibrator manifest, and primary RQs. Gate failure suppresses all Identity-side reporting. Gate passage permits only a separately labeled offline exploratory annotation after primary inference; it cannot change the EDR/NDR candidate universe or $K = 2$ contract.

E.4 Ground Truth Store and Offline Join

The Ground Truth Store uses a separate database role. Join keys are run ID, harmless marker ID, target entity namespace, and versioned time interval. The Evaluation Harness first verifies that online inference and `AuditRecord` insertion have completed, then performs the join under the frozen label-policy digest. A two-reviewer adjudication resolves disagreements; unresolved cases become ambiguous rather than being forced into the binary cohort.

Offline matching first requires temporal IoU at least .10, entity Jaccard at least .50, and one exact harmless marker or registered action. Within one run, maximum-weight bipartite assignment uses $w = .5A_{cov} + .3J_{entity} + .2IoU_{time} \geq \theta_{match} = .50$; ties use entity overlap, absolute time difference, then lexical IDs. A revision is positive when malicious-action coverage is at least $\theta_{mal} = .50$ and unrelated benign contamination is below $\theta_{contam} = .25$; mixed when both thresholds are met; negative when malicious coverage is zero and activity has registered benign/background attribution; all residual, missing, or conflicting cases are ambiguous. Labels are revision specific and join only after `AuditRecord` insertion. Sensitivity varies $\theta_{mal} \in \{.25, .50, .75\}$, $\theta_{contam} \in \{.10, .25, .50\}$, and $\theta_{match} \in \{.40, .50, .60\}$ without altering the primary registry after seeing performance.

E.5 Leakage Tests

Continuous schema tests inject each forbidden field into every online object and require a hard rejection. Partition tests verify that

every descendant of one run has exactly one split. Feature-lineage tests prove that all scorer inputs trace to allowed online fields. SQL-role tests deny online reads from Ground Truth Store tables. A canary scenario token is placed only in the offline store and must be absent from every normalized event, revision, feature, model input, and AuditRecord. Any failure invalidates the affected run and blocks locked evaluation.

F EXPERIMENTAL CONTRACT AND REPRODUCIBILITY

F.1 Split and Freeze Manifest

The registered views prevent revision multiplicity from changing a claim. C_{form} contains one terminal revision per surviving lineage at $T_{form} = t_{run_end} + L_{wm} + W_{rev} = t_{run_end} + 1020$ s, after removing merged parents. C_{dec} contains the first finalized, decision-eligible revision per terminal episode and is the only primary RQ2 population. C_{stream} contains every AuditRecord keyed by $(episode_id, revision)$ and is used only for RQ3/RQ4 transitions and cost.

The independent split unit is a base campaign, stratified by scenario family and benign mode. Within each stratum, campaigns sort by ascending SHA-256 of 20260710:base_campaign_id. In each consecutive 20-run cycle, ranks 0–9 are training, 10–13 development, 14–16 calibration, and 17–19 locked test, targeting 50/20/15/15 percent. Replay and outage descendants inherit partition and bootstrap cluster; any mismatch invalidates the complete cluster. The registry requires run, base, parent, and pair IDs; scenario; benign mode; seed; partition; cluster; role; and manifest digest. The freeze manifest additionally binds code, container/image, formation, feature, normalizer, scorer, calibrator, label, baseline, intervention, metric, bin, and analysis-script digests. The machine policy is artifacts/contracts/split_policy.v1.json; no post-label manual balancing is permitted.

F.2 Baseline Implementation Registry

EB1 uses non-overlapping event-time windows of width W_{gap} with no entity or causal merge. EB2 adds the requirement that consecutive records share one specific, non-generic entity. EB3 is Algorithm 1; EB3-H changes only hub suppression. SB1 is the maximum source-normalized vendor severity among eligible episode records. SB2 fits the same constrained logit procedure as Equation 7 with $w_C = 0$. SB3 is the proposed raw logit. SB4 reports the uncalibrated sigmoid score $\sigma(z)$ and is not called a probability. SB5 applies the global affine calibrator. SB6 is the conditional group mapping. SB7 maps an unavailable source’s absent detector value to zero and ignores manifest support. OB1 is the physical-outage-only available-source denominator and returns no corroboration when fewer than two source families are healthy. No entry has an alternative max/mean, weight set, model capacity, or post-test implementation.

F.3 Frozen Perturbation Grid and Outage Procedure

Every replay child seed is HMAC-SHA256 over base-run ID, intervention, and level with key 20260710; descendants retain the base label, split, and cluster. The exact registry is artifacts/contracts/perturbations.v1.json.

Physical outage uses the same scenario/seed pair as a normal run. After a 120-second steady state, the selected EDR or NDR collector is stopped or network-isolated for 120 or 600 seconds, then restored. The procedure records last healthy heartbeat, first degraded/down SourceCoverage interval, watermark movement, emitted revisions, κ , eligibility, recovery heartbeat, and stable

revision. Recovery requires two healthy heartbeats, a valid parser, backlog no greater than 120 s, and no new revision for 900 s. Safety aborts the run if isolation affects the control plane or a non-laboratory route.

F.4 Matching and Metric Formulas

Before primary assignment, a C_{form} terminal revision is match eligible only with temporal IoU at least .10, entity Jaccard at least .50, and one exact marker or registered action. For eligible pairs,

$$w = .5A_{cov} + .3J_{entity} + .2IoU_{time}; \quad w \geq \theta_{match} = .50. \quad (13)$$

Maximum-weight bipartite assignment operates within one run; ties use greater entity overlap, smaller absolute time difference, then lexical IDs. Formation metrics use C_{form} , ranking and calibration use C_{dec} , and only transition/cost analyses use C_{stream} .

For ground-truth malicious-action set A and the actions covered by assigned revisions $\hat{A}$,

$$\text{ActionCoverage} = \frac{|A \cap \hat{A}|}{|\hat{A}|}. \quad (14)$$

For run r , let G_r be its ground-truth action groups and n_g the number of distinct terminal predicted episodes assigned to group g . Fragmentation is $|G_r|^{-1} \sum_{g \in G_r} \max(0, n_g - 1)$. Over-merge is the fraction of terminal predicted episodes assigned to more than one distinct action group in the same run; the unit is never a mutually exclusive run ID. Benign contamination is the fraction of unrelated benign events inside revisions assigned to malicious groups. Episode precision, recall, and F1 use the locked one-to-one assignment.

For supported records $i = 1, \dots, n_{sup}$,

$$\text{Brier} = \frac{1}{n_{sup}} \sum_i (P_i - Y_i)^2, \quad (15)$$

$$\text{NLL} = -\frac{1}{n_{sup}} \sum_i [Y_i \log P_i + (1 - Y_i) \log(1 - P_i)]. \quad (16)$$

Out-of-support records are absent from both sums. Reliability uses ten fixed equal-width bins on $[0, 1]$ and always reports bin counts. Supported decision coverage is $|C_{sup}|/|C_{dec}|$. Ranking reports AUPRC and recall among the top $\max(1, \lceil 0.05|C_{dec,r}| \rceil)$ revisions in each run; AUROC and precision at the same budget are secondary. All primary summaries are run-macro means; pooled micro results are supplemental.

F.5 Run-Level Uncertainty and RQ Matrix

Primary intervals use 2,000 cluster-bootstrap resamples of complete campaign/run clusters with seed 20260710 and percentile endpoints at 2.5% and 97.5%. For paired replay and outage contrasts, the resampled object is the base-run pair. A resample lacking both classes for a class-dependent metric is marked undefined for that metric and counted; it is not replaced by zero. Run-level p50/p95/p99 latency and throughput are computed before cross-run aggregation.

Campaign-level splits, artifact freeze, and one-pass evaluation

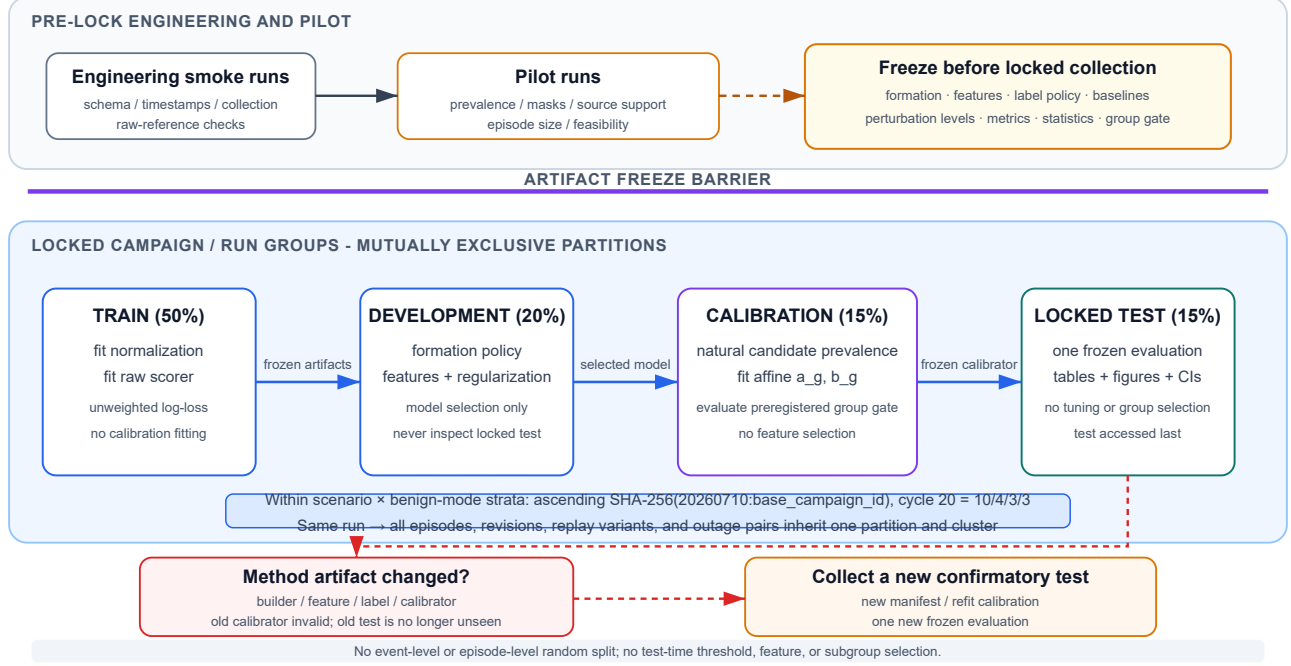


Figure 9: Campaign-level split and freeze workflow. Deterministic 50/20/15/15 assignment is performed before label release, and all descendants inherit the base campaign partition and bootstrap cluster.

Table 15: Pre-registered replay perturbations. Each variant includes an unchanged control and uses a deterministic child seed derived from the base-run ID.

ID	Intervention	Frozen levels	Invariant
RP0	unchanged replay	no mutation	exact canonical-hash equality required
RP1	source-event mask	EDR only; NDR only	SourceCoverage unchanged
RP2	independent random drop	5%; 10%; 25%	retained event times and health unchanged
RP3	bounded burst drop	30 s; 120 s; 300 s	one seeded interval per run
RP4	ingest delay	30 s; 120 s; 600 s	event time unchanged
RP5	arrival reorder	buffer of 30 s; 120 s; 600 s	payload and event time unchanged
RP6	event-time clock skew	−120; −30; +30; +120 s	ingest time and health unchanged
RP7	dependency-aware duplicate	5%; 10%; 25%	duplicate retains root dependency group

F.6 Result Registry and Canonical Replay

RT1 contains split, population, run, candidate, positive, negative, mixed, ambiguous, and prevalence counts. RT2 contains formation metrics. RT3a contains ranking comparators; RT3b contains supported calibration only. RT4 contains paired replay effects. RT5 contains physical outage and recovery, including OB1. RT6 contains determinism and resource cost. RT7 contains registered error categories. Every unexecuted cell is serialized as pending, never zero; N/A is reserved for structurally inapplicable fields. The machine schema is `artifacts/contracts/result_registry.v1.json`.

The RQ4 canonical payload consists of episode and revision IDs, ordered event membership, link reasons, $D, C, a^{det}, z, P, \kappa, U$, detector and health states, `decision_eligible`, versions, policy and artifact hashes, and evidence references. Equality is byte-for-byte

SHA-256 equality under Appendix A.3. Execution timestamps and host-specific runtime metadata are excluded from this claim.

The formal machine contracts and this expanded appendix are packaged as `TrustEpisode_reproducibility_companion_v1.zip`; the adjacent `.sha256` sidecar authenticates the exact distributed bytes. The bundle contains no fabricated runs or results.

Table 16: Frozen RQ-comparison-metric matrix.

RQ	Comparison/intervention	Primary outcomes	Unit
RQ1	EB1, EB2, EB3, EB3-H	action coverage; fragmentation; over-merge	campaign/run
RQ2	SB1-SB5; SB6 conditional	AUPRC; fixed-budget recall; Brier; NLL; reliability	campaign/run
RQ3	paired replay grid; separate outage pairs	formation/z/support/U/revision changes; outage detection/recovery	base-run pair
RQ4	unchanged replay; frozen hardware run	canonical digest equality; throughput; latency; memory; storage; replay time	campaign/run

Table 17: RT3a ranking-result shell. SB1 has no probability mapping.

Scorer	AUPRC $\uparrow$	Recall@budget $\uparrow$	AUROC $\uparrow$	Precision@ k $\uparrow$
SB1 static severity	pending	pending	pending	pending
SB2 D only	pending	pending	pending	pending
SB3 $D + C$ raw	pending	pending	pending	pending

Table 18: RT3b calibration-result shell over supported records only.

Mapping	Brier $\downarrow$	NLL $\downarrow$	Reliability	Supported coverage	Out-of-support rate
SB4 uncalibrated sigmoid score	pending	pending	pending	pending	pending
SB5 supported global	pending	pending	pending	pending	pending
SB6 supported group	pending	pending	pending	pending	pending

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